A Post-Fabrication-Tunable Time-Domain Multi-Row Read for RRAM Processing-in-Memory

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***Abstract*—Resistive RAM (RRAM) turns a memory array into a computing element, because a read that opens several rows sums the conductances of the selected cells on the bit line. The circuit that resolves this sum is designed against a device model, yet the compact models of RRAM processes capture neither the resistance spread of a programmed array nor the capacitance of the memory stack, so a read designed for one assumed operating point can miss the point at which the fabricated cells actually sit. This paper presents a read path whose operating point is chosen after fabrication. Each column converts the bit-line discharge time into a thermometer code with a delay-line time-to-digital converter, and the sampling window, tap pitch, and drive levels are set from registers and supply pads, so that opening several rows turns a read into an in-memory NOR or majority without a reference cell. The read path and its controller are implemented in a tape-out-ready 64×64 one-transistor-one-resistor macro. Post-layout simulation with the extracted bit-line loading separates a single row by all eight taps and resolves an 8-row NOR by six taps and an 8-row majority by two, in 0.45 µs and 64 to 107 pJ per 64-bit access, and a Monte Carlo analysis converts these margins into the resistance spread the device has to meet, which the single-row read satisfies at the edge of the characterized high-resistance state and the multi-row logic reaches with a deeper reset.**

***Index Terms*—RRAM, 1T1R, processing-in-memory, time-domain readout, time-to-digital converter.**

I. Introduction

Resistive RAM (RRAM) stores data as the conductance of a two-terminal device, and in a one-transistor-one-resistor (1T1R) array this conductance can be read in parallel. When several word lines open at once, the bit line sums the conductances of the selected cells, and the array becomes a computing element rather than a store. A multi-row read then yields bulk bitwise operations [1] or an analog multiply-accumulate (MAC) [2], [3] in a single access [4], and stateful memristive logic complements it by writing the result back into a cell as a material implication [5], a NOR [6], or a majority [7]. Silicon demonstrations of this processing-in-memory (PiM) have relied on dedicated sense circuits with tracking references or calibration [8], [9], and time-domain readouts with time-to-digital converters (TDCs) have digitized in-memory MAC compactly [10], [11].

Every one of these read circuits is designed against a device model, and this is where RRAM departs from a mature memory. The compact model of an RRAM module is fitted to a few programmed states, and it captures neither the resistance spread of a verify-programmed array nor the capacitance of the memory stack itself, while the fabricated arrays differ from the model, and from one another, in both. On the HfAlO module used here, the high-resistance state lands anywhere between 60 and 220 kΩ depending on the reset verify, its spread across devices is 0.5 to 1 in $\ln R$ [12], [13], and the 22:1 window between the two states reaches the bit line as a 3.6:1 current contrast once the access transistor sits in series (Section IV). A reference-tracked sense amplifier compares against reference cells programmed with the same uncertain device, and a calibration loop still has to be designed before the devices are characterized. What remains open is a read whose decision point does not come from the model at all, but is placed after the fabricated cells have been measured.

This work proposes a time-domain read with exactly that property. Instead of a voltage-mode sense amplifier, every column converts the time the bit line takes to discharge through the selected cells into a thermometer code with an eight-tap delay-line TDC, and the sampling window, the tap pitch, and the drive levels are set by registers and supply pads rather than by transistor sizing. Because the decision is a threshold on a code, opening several rows turns the same read into an in-memory NOR or majority. The read path and its controller are implemented in a tape-out-ready 64×64 1T1R macro. Post-layout simulation with the extracted bit-line loading separates a single row by all eight taps and resolves an 8-row NOR by six taps and an 8-row majority by two, with the cells biased below the 0.2 V read condition of the device, and an access returns 64 decisions in 0.45 µs for 64 to 107 pJ. A Monte Carlo analysis on these margins then turns the evaluation into a budget for the device, namely the resistance spread each operation tolerates, which the single-row read meets at the edge of the characterized high-resistance state and the multi-row logic meets with a deeper reset.

II. Post-Fabrication-Tunable Time-Domain Read

Fig. 1 shows the read built for cells whose operating point the model cannot predict. A conventional sense amplifier asks whether the bit line is above or below a reference at one instant, whereas this read asks how long the bit line takes to discharge. A read precharges the bit line BL to the drive voltage V_{drain}, grounds the source line SL and opens the selected rows, so that BL discharges through the selected cells with a time constant set by the cell resistance, the access-transistor resistance, and the bit-line capacitance. A buffer with a threshold of about 0.6 V converts BL into a digital signal, and eight flip-flops sample this signal on the eight taps of a delay line that a

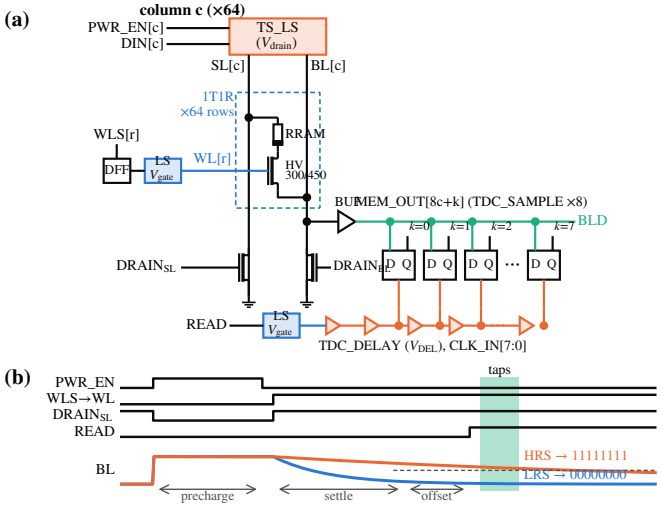


Fig. 1. (a) One column of the macro with its TDC. (b) Read sequence: the source line is grounded as the rows open, and the eight-tap window (shaded) samples the discharge after the offset.

READ pulse triggers. The result is a thermometer code that counts the taps at which BL was still high, so that a low-resistance (LRS) cell gives a short code and a high-resistance (HRS) cell a long one (Fig. 1(b)).

Three knobs place the decision, and none of them is a transistor size. The controller inserts a signed offset between its drain step and the READ pulse in 10 ns steps, positive when the drain comes first, which moves the sampling window along the discharge; a dedicated supply pad V_{DEL} sets the delay of the line and with it the tap pitch; and the word-line supply V_{gate} sets the access-transistor resistance together with the level-shifter delay. The first tap fires 3.4 ns after the READ edge, and the pitch is 0.35, 0.24, and 0.18 ns at $V_{\mathrm{DEL}} = 0.8$, 0.9, and 1.0 V, so the window spans 2.5 to 1.3 ns; below about 0.7 V the swing of the line no longer clocks the 1.2 V sampling flip-flops, which bounds the window to about 5 ns, narrower than the step in which it moves. All three are registers or pads, so the window is placed after the fabricated cells have been measured, with no reference cell programmed from the same uncertain device.

With n rows open, the column conductance is the sum of the selected cells, so the discharge time, and with it the code $c(k)$, falls monotonically with the number k of LRS cells among them. The in-memory operations are thresholds on this code, each with a margin measured in taps. A NOR over the n rows separates $k = 0$ from $k \geq 1$ with a margin of $c(0) - c(1)$, and a majority separates $k \leq n/2$ from $k > n/2$ with a margin of $c(n/2) - c(n/2+1)$.

III. Macro and Controller

The read path is implemented in a 64×64 1T1R macro in the IHP SG13G2 130 nm BiCMOS process, whose back-end HfAlO memristor (MEMRES) module [12] provides the memory devices (Fig. 2). The array holds 4096 cells on a 2.26×2.77 µm pitch, each a high-voltage (HV) nMOS access

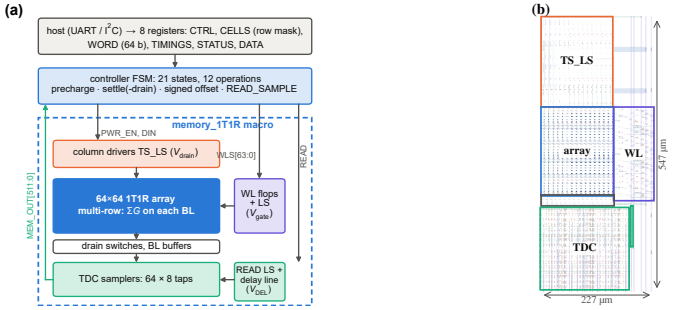


Fig. 2. (a) Macro and controller architecture. (b) Tape-out-ready layout, 226.5×547.4 µm.

transistor (W/L = 300/450 nm) in series with a MEMRES device whose top electrode connects to the source line of its column (Fig. 1(a)). Each column has a tri-state level-shifting driver supplied from V_{drain} that drives SL or BL high for the writes and for the read precharge, and two global HV switches ground all source lines or all bit lines. Each word line is driven through a pipeline flip-flop and a cross-coupled level shifter supplied from V_{gate}, so that an arbitrary 64-bit row mask written in one cycle opens any subset of rows in the next. The delay line is shared by all 64 columns and supplied from the V_{DEL} pad, and the 512 sampled bits return as the result of an access.

A 21-state finite-state machine (FSM) drives the macro with 12 operations: reads, SET, RESET, and mixed-polarity writes of a 64-bit word into the selected rows, a discharge-length test, and a sample capture of sixteen codes along one discharge. A read precharges for a programmable number of cycles, opens the row mask for a settle cycle with the source lines grounded, and then applies the drain step and READ with the signed offset of Section II. Eight 32-bit registers are written from a host over a serial link, and a host procedure searches V_{gate}, V_{DEL}, V_{drain}, the settle time, and the offset for the operating point that best separates the classes of each operation, which is the procedure Section IV follows. The macro is full custom and the controller is synthesized and placed around it; in an analog/mixed-signal simulation of the controller RTL against the macro netlist the sequenced reads return the codes of the stimulus-driven runs, and the design signs off at 100 MHz in the slow corner with clean DRC, antenna, and LVS.

IV. Evaluation

A. Simulation setup

The read is evaluated on an 8×8 slice of the layout netlist of the macro, with its drivers, word-line flip-flops and level shifters, drain switches, delay line, and TDC flip-flops. The slice is simulated with the foundry's PSP transistor models at the typical corner; process corners and temperature follow in Section IV-D. The cells are linear resistors, 9 kΩ for the LRS and 200 kΩ for the HRS, the top of the range IHP has characterized (1 µA at 0.2 V), with 1 MΩ added as the deep-reset state that a stronger reset verify would approach. At the

TABLE I
BEST OPERATING POINT PER OPERATION (SLICE, LRS 9 kΩ)

op.	HRS (Ω)	$V_{drain}/V_{gate}/V_{DEL}$ (V)	off. (ns)	codes $c(k)$ $k = 0 \ldots 8$	margin (taps)
read, 1 row	200 k	0.8/0.8/0.8	20	8, 0 (HRS, LRS)	8
NOR, 8 rows	200 k	0.9/0.8/1.0	10	7 1 0 0 0 0 0 0 0 0	6
majority, 8 rows	200 k	1.0/0.8/1.0	10	8 8 8 6 3 1 0 0 0	2
NOR, 8 rows	1 M†	0.8/0.8/0.9	10	8 0 0 0 0 0 0 0 0	8
majority, 8 rows	1 M†	1.0/0.8/1.0	10	8 8 8 8 7 3 0 0 0	4

direct transients, C_{MIM} = 10 fF, extracted line capacitance; †deep-reset state; V_{RRAM}, I_{cell}: largest cell voltage and current over the row's runs (different cells and instants).

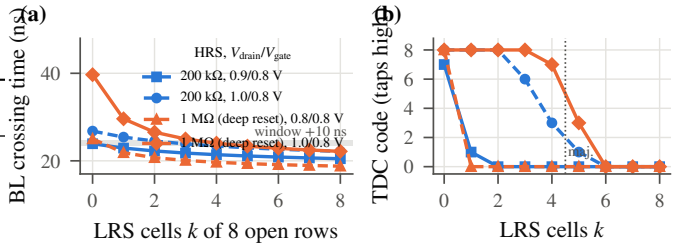


Fig. 3. 8-row read with k cells in the LRS, at 200 kΩ and 1 MΩ HRS: (a) bit-line crossing time, with the +10 ns window shaded; (b) simulated TDC code.

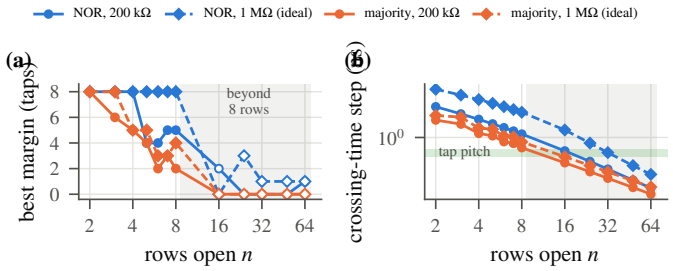


Fig. 4. Rows per access: (a) best NOR and majority margin; (b) crossing-time step behind each decision against the tap pitch (shaded). Hollow markers: at the best 8-row supplies.

0.06 to 0.12 V the read places across a cell, the HRS non-linearity of these devices raises R by about 10 % over its 0.2 V value and the LRS is ohmic [14], and at the two states the IHP compact model can produce (1.4 and 12 kΩ) it reproduces the crossing times of the resistor runs within 0.06 ns. The supplies are confined to $V_{drain}, V_{gate}, V_{DEL} \in \{0.8, 0.9, 1.0\}$ V so that a read never approaches a write condition.

A resistor cell omits two capacitances that set the discharge, and both are added explicitly. The first is the line loading, for which an RC extraction of the full macro gives 47 to 62 fF of interconnect on each bit line and 21 to 36 fF on each source line, so the slice adds 70 fF to BL and 30 fF to SL for the wire and source junctions of the 56 rows it does not instantiate. The second is the MEMRES stack itself, a metal-insulator-metal capacitor that neither the compact model nor the extraction contains. A Sentaurus solution of the drawn stack (0.36 µm², 6 nm of HfAlO, ε_r = 12 to 20) gives C_{MIM} = 6.8 to 11.0 fF per cell, and all results use 10 fF. This capacitance bypasses a selected cell with a time constant of 2 ns at 200 kΩ and 10 ns at 1 MΩ, the scale of the differences the read resolves. The level-shifting driver charges the loaded bit line slowly at V_{drain} = 0.8 V, where the default 10-cycle precharge leaves BL at 0.49 V, so all runs precharge for 40 cycles (400 ns).

An access takes the precharge, the settle, the offset, and the READ pulse, about 0.45 µs for 64 decisions at 100 MHz, and draws 8 to 13 pJ per slice from the supply currents, or 64 to 107 pJ per access scaled to the macro with the controller excluded (1.0 to 1.7 pJ per 64-bit result). Of this, 53 to 69 % is the clock energy of the word-line flip-flops during the precharge and, at V_{drain} = 0.8 V, 21 % the static current of the bit-line buffers whose input sits at V_{drain}, while the discharge of the array and the delay line take 0.5 to 1.1 pJ; a clock gate during the precharge would bring an access to 21 to 50 pJ.

B. Single-row read

At V_{gate} = 0.8 V the access transistor is in weak inversion and limits the cell current, in a DC sweep with 0.9 V on BL, to 0.5 to 1.7 µA for 200 to 9 kΩ, so the 22:1 resistance ratio of the cells reaches the bit line as a current contrast of only 3.6:1 (2.5:1 fitted to the transients). Even so, a single row separates HRS from LRS by the full eight taps (Table I). With all supplies at 0.8 V the 200 kΩ cell crosses the buffer threshold 50.7 ns after the drain step and the LRS cell after 28.2 ns, a window at a +20 ns offset lies between them, and the cell sees at most 0.09 V and 1.3 µA.

C. Multi-row NOR and majority

Because V_{drain} stays at or below 1 V, the settle cycle must ground SL (a controller bit) since with SL floating, charge sharing takes BL below the buffer threshold before READ. The full supply grid was simulated for both HRS values, the codes at every window position were computed from the simulated crossing times and the tap tables, and the best point of each operation was then simulated directly. Table I and Fig. 3 show these points; the computed codes are one tap conservative, so the 8-row NOR margin is six taps in the direct transient and five in the crossing-time model behind Fig. 4 and the temperature results.

At 200 kΩ every multi-row operation is resolved at V_{gate} = 0.8 V and at no higher word-line voltage, because the current-limited cells stretch the all-HRS crossing to 21 to 27 ns (Fig. 3(a)), far enough for V_{drain} to place the band against the window grid, whereas at $V_{gate} \geq 0.9$ V the band is shorter than the 10 ns offset step and falls between window positions. At V_{drain} = 0.9 V the all-HRS crossing (23.9 ns) lies inside the window and the $k = 1$ crossing (22.9 ns) before it, which gives the codes 7, 1, 0, … of Fig. 3(b) and a NOR margin of six taps, and the deep-reset 1 MΩ reaches the full scale. The majority is harder, because the step from k to $k+1$ shrinks as $1/k^2$, so $c(4) - c(5)$ needs the window on the compressed part of the band; V_{drain} = 1.0 V moves it there and gives a majority by two taps, or four at 1 MΩ. The 10 ns step, one clock period, is thus the main limitation of the design, since it caps both the row count and the hot-corner margins that follow.

D. Rows per access, corners, and temperature

Fig. 4 extends the same model from one to eight rows over the full supply grid, and to 16 to 64 rows at the best 8-row supplies. At 200 kΩ the NOR keeps all eight taps up to four rows and four to five taps up to eight, while the majority falls from eight taps at two rows to two or three at six to eight. Every open row adds an HRS current against which one LRS cell must stand out, so the crossing-time step behind each decision falls roughly as $1/n^2$ (Fig. 4(b)), from 17 ns (NOR) and 5 ns (majority) at two rows to 1.4 and 0.4 ns at eight. From 16 rows on, the step at 200 kΩ is at or below a tap pitch. One access therefore decides over at most eight rows at the characterized HRS, and a wider NOR is split into eight-row groups that the host combines.

Over the five PSP corners and −40 to 125 °C, with the cell resistances scaled by the measured temperature coefficients [14], the all-HRS crossing of the 8-row read moves from 16 ns (ff, 125 °C) to 60 ns (ss, −40 °C) against the 10 ns window step, and this is where the tunability matters. With the room-temperature setting held fixed, the single-row read keeps its eight taps everywhere but the 8-row operations are lost at both temperature extremes; re-searching the offset decides the single-row read in 14 of the 15 corner-temperature pairs, and re-tuning V_{drain}, V_{gate}, V_{DEL}, and the offset per temperature recovers the 8-row NOR at −40 °C with five taps and the majority with two, the margins of 27 °C, but not at 125 °C, where the band is shorter than the step and one tap remains.

E. Read disturb and the device budget

A read biases the selected cells in the RESET polarity, with BL high and SL grounded, so repeated reads push the cells toward the HRS. At the best points a cell sees at most 0.06 to 0.12 V and 0.8 to 1.3 µA, below the 0.2 V read condition of the device and a sixth of its 8 µA RESET target. On these devices 10^5 pulses of 10 µs at 0.1 to 0.2 V leave every level unchanged and drift starts at 0.3 V, while retention alone lowers the LRS by 15 % in 100 h at room temperature [15], so the refresh interval is set by retention and temperature rather than by the read count.

The margins above hold at the typical corner with fixed resistances, and the final question is how much spread they tolerate. A Monte Carlo draws every open cell from a lognormal distribution ($\sigma_{\ln R} = 0.1$ for the LRS, swept for the HRS) behind the 0.08 MΩ access resistance of the weak-inversion transistor (a 3:1 contrast), interpolates the simulated crossing times in the summed conductance, and counts the errors of one column decision at an ideal threshold midway between the nominal sums (10^5 draws); 160 SPICE runs with drawn resistances reproduce its crossing-time spread. At a 10^{-3} error per decision the HRS spread is limited to 0.45 for a single row, 0.33 for two rows, and 0.17 for the 8-row NOR (0.24 for the majority), close to $0.46/\sqrt{n}$, the Gaussian-sum form in which each open HRS cell adds its own noise. The module's HfAlO stack is that of IHP's 250 nm arrays, where the verify-programmed HRS shows a spread of 0.5 with its conductive edge clipped by the verify at 40 kΩ [12]; the Al-doped 130 nm devices show 0.5 to 1 with a median of 90 to 220 kΩ and the edge pinned at 50 kΩ [13]. The single-row read therefore sits at the edge of the characterized state, with the clipped edge in its favor, whereas the 8-row logic needs the deeper reset of the 1 MΩ rows of Table I, which buys a full-scale 8-row NOR, a four-tap majority, and an 8-row spread limit of 0.5 (0.6 for the majority). A deeper reset, a tighter verify window, and fewer rows per access are the levers (a higher word-line voltage adds only 30 to 45 % of contrast), and the first two are programming-time costs: a deeper reset takes more pulses of the incremental step pulse with verify algorithm (ISPVA), about twelve of 12 µs [12], and with Al doping leaves a quarter of the cells above 1 MΩ at the read-current floor [13].

TABLE II
COMPARISON WITH RRAM AND TIME-DOMAIN MACROS

	node	memory	readout	cells/access	reported metric
[8]	65 nm	1 Mb 1T1R	current SA, 3 b	9–25	14.8 ns/cycle
[9]	130 nm	3 Mb 1T1R	ADC, calibrated	256	43 TOPS/W (1 b)
[10]	40 nm	none on chip	counter TDC	25–150, serial	12.1 TOPS/W
[11]	180 nm	8 kb SRAM	counter TDC	32 per chain	0.3–0.8 TOPS/W[a]
this work	130 nm	4 kb 1T1R	8-tap TDC, tunable	≤ 8	1.0–1.7 pJ/result[b]

[a] 8×8 b at 1.2 V; the title's 768–2124 TOPS/W is normalized to 28 nm and 1 b. [b] per 64-bit result, 8×8 slice post-layout SPICE, 0.45 µs per access; SA: sense amplifier.

F. Comparison and limitations

Table II places the read next to the silicon macros it draws on. Those macros are MAC engines, so the table compares structure rather than figures of merit, and against their reference-tracked or calibrated readouts this work trades throughput for a bitwise read of up to eight rows whose every timing and drive level is a register or a pad. The evaluation rests on a slice with resistor cells and one C_{MIM}, the Monte Carlo varies the cell resistances only, and the disturb bound rests on SET-polarity data, so the HRS spread, the access-transistor mismatch, and the RESET-polarity disturb are the first quantities to measure on silicon, and a fractional-cycle offset is the first change for a second version.

V. Conclusion

The read circuit of an RRAM PiM array is designed against a model the fabricated cells do not follow, so this paper placed its operating point after fabrication: the discharge time becomes a code whose window registers and supply pads move, and several open rows turn the read into a NOR or majority. In a tape-out-ready 64×64 1T1R macro with its extracted loading, the read separates a single row by eight taps and resolves an 8-row NOR by six taps and a majority by two, and re-tuning recovers the single-row read at every temperature and the 8-row NOR in the cold, where a fixed setting loses both. The same simulation yields the budget the device has to meet, a resistance spread of 0.45 for a single row and 0.17 for eight rows at the characterized high-resistance state, or 0.5 with a deeper reset, which is the first quantity to measure on silicon. A read that is tuned rather than assumed survives the one thing a research device is certain to do, which is to come back different from its model.

References

[1] S. Li, C. Xu, Q. Zou, J. Zhao, Y. Lu, and Y. Xie, "Pinatubo: A processing-in-memory architecture for bulk bitwise operations in emerging non-volatile memories," in *Proc. 53rd Annu. Design Autom. Conf. (DAC)*, 2016, pp. 1–6.

[2] P. Chi, S. Li, C. Xu, T. Zhang, J. Zhao, Y. Liu, Y. Wang, and Y. Xie, "PRIME: A novel processing-in-memory architecture for neural network computation in ReRAM-based main memory," in *Proc. ACM/IEEE 43rd Annu. Int. Symp. Comput. Archit. (ISCA)*, 2016, pp. 27–39.

[3] A. Shafiee, A. Nag, N. Muralimanohar, R. Balasubramonian, J. P. Strachan, M. Hu, R. S. Williams, and V. Srikumar, "ISAAC: A convolutional neural network accelerator with in-situ analog arithmetic in crossbars," in *Proc. ACM/IEEE 43rd Annu. Int. Symp. Comput. Archit. (ISCA)*, 2016, pp. 14–26.

[4] D. Ielmini and H.-S. P. Wong, "In-memory computing with resistive switching devices," *Nature Electron.*, vol. 1, no. 6, pp. 333–343, 2018.

[5] J. Borghetti, G. S. Snider, P. J. Kuekes, J. J. Yang, D. R. Stewart, and R. S. Williams, "'Memristive' switches enable 'stateful' logic operations via material implication," *Nature*, vol. 464, no. 7290, pp. 873–876, 2010.

[6] S. Kvatinsky, D. Belousov, S. Liman, G. Satat, N. Wald, E. G. Friedman, A. Kolodny, and U. C. Weiser, "MAGIC—memristor-aided logic," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 61, no. 11, pp. 895–899, 2014.

[7] P.-E. Gaillardon, L. Amarú, A. Siemon, E. Linn, R. Waser, A. Chattopadhyay, and G. De Micheli, "The programmable logic-in-memory (PLiM) computer," in *Proc. Design, Autom. Test Eur. Conf. Exhib. (DATE)*, 2016, pp. 427–432.

[8] W.-H. Chen, K.-X. Li, W.-Y. Lin, K.-H. Hsu, P.-Y. Li, C.-H. Yang, C.-X. Xue, E.-Y. Yang, Y.-K. Chen, Y.-S. Chang, T.-H. Hsu, Y.-C. King, C.-J. Lin, R.-S. Liu, C.-C. Hsieh, K.-T. Tang, and M.-F. Chang, "A 65nm 1Mb nonvolatile computing-in-memory ReRAM macro with sub-16ns multiply-and-accumulate for binary DNN AI edge processors," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, 2018, pp. 494–496.

[9] W. Wan, R. Kubendran, C. Schaefer, S. B. Eryilmaz, W. Zhang, D. Wu, S. Deiss, P. Raina, H. Qian, B. Gao, S. Joshi, H. Wu, H.-S. P. Wong, and G. Cauwenberghs, "A compute-in-memory chip based on resistive random-access memory," *Nature*, vol. 608, no. 7923, pp. 504–512, 2022.

[10] A. Sayal, S. S. T. Nibhanupudi, S. Fathima, and J. P. Kulkarni, "A 12.08-TOPS/W all-digital time-domain CNN engine using bi-directional memory delay lines for energy efficient edge computing," *IEEE J. Solid-State Circuits*, vol. 55, no. 1, pp. 60–75, 2020.

[11] C. Xue, X. Qiao, X. Ren, G. Du, Y. Wang, and Y. He, "A 768.7–2124.2 TOPS/W time-domain computing-in-memory macro with low static leakage and precision-configurable TDC," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 71, no. 5, pp. 2789–2793, 2024.

[12] E. Perez, A. Grossi, C. Zambelli, P. Olivo, R. Roelofs, and C. Wenger, "Reduction of the cell-to-cell variability in $Hf_{1-x}Al_xO_y$ based RRAM arrays by using program algorithms," *IEEE Electron Device Lett.*, vol. 38, no. 2, pp. 175–178, 2017.

[13] A. Baroni, E. Perez, K. Dorai Swamy Reddy, S. Pechmann, C. Wenger, D. Ielmini, and C. Zambelli, "Enhancing RRAM reliability: Exploring the effects of Al doping on HfO_2-based devices," *IEEE Trans. Device Mater. Reliab.*, vol. 25, no. 3, pp. 379–387, 2025.

[14] T. Zanotti, T. Rizzi, E. Perez-Bosch Quesada, A. Baroni, K. Dorai Swamy Reddy, E. Perez, C. Wenger, P. Pavan, and F. M. Puglisi, "Physics-based RRAM compact model for multilevel programming across multiple timescales," *J. Semicond.*, vol. 47, no. 6, p. 062301, 2026.

[15] A. Baroni, J. Wen, E. Perez, K. Dorai Swamy Reddy, M. Uhlmann, S. Pechmann, C. Wenger, D. Ielmini, and C. Zambelli, "Field- and temperature-induced drift in multi-level Al-doped HfO_2 RRAM," *IEEE Trans. Device Mater. Reliab.*, 2026, early access.